CLADE: Category-Level Answering from Deficient Evidence in Retrieval-Augmented Language Models

Anonymous submission

Abstract

Retrieval-augmented systems often hand a model a page that is true and relevant but names only the family of the answer: the question asks which disease a drug treats, and the page says "hyperlipidemia" where the answer is "familial hyperlipidemia." Work on sufficient context treats such pages as simply insufficient. We argue they form a distinct case, *resolvable insufficiency*: the text cannot settle the answer, but its family name indexes a short candidate list in an ontology. Models meet it with CLADE (category-level answering from deficient evidence): they answer instead of declining or looking further. On 856 PrimeKG questions, a family-only page weakens the push toward "not enough information" by 13.8 points relative to a silent page (pooled over three 7–8B models, $p = 4 \times 10^{-5}$; significant for two of the three individually), and the freed answers are about as often wrong as right. A counterfactual edit isolates the cause: replacing the disease name with its family name lowers declines by 17.4 points relative to deleting the sentence. Confidence (AUROC 0.59–0.72 on family-only pages, against 0.73–0.87 on named pages), a 14B model (58% precision, against 96%) and an instruction to decline (more declines on every kind of page) all leave CLADE in place. It replicates on species interactions from GloBI, with a taxonomy as the symbolic structure (900 questions; decline effect -14.3 points, $p = 10^{-9}$), where the freed answers are mostly wrong. Resolving the insufficiency is only partly solved: the ontology list holds the answer 98.5% of the time once the family is known, but the family is found from the page only 19.8% of the time, and a popularity prior beats both the model and literature evidence at choosing within it. The results locate where a neural reader should hand over to symbolic knowledge, and which steps of that handover remain open.

1 Introduction

A retrieval-augmented system answers a question by reading a page (Lewis et al. 2020). Much recent work asks what the model should do when the page is not enough. The usual framing is binary: the context is sufficient or it is not (Joren et al. 2025), and a well-behaved model answers in the first case and abstains in the second (Wen et al. 2025).

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Many pages fall between these cases in a structured way. Consider the question "What does lovastatin treat?" with the answer *familial hyperlipidemia*, and a page that says "lovastatin is used to treat hyperlipidemia" (Figure 1). The page is true and relevant. It rules out every disease that is not a hyperlipidemia. It does not say which one. A model can answer anyway, say "not enough information," or look further. Looking further has a natural symbolic form: a disease ontology lists the members of the family, and each can be checked.

We call this case *resolvable insufficiency*. The page is insufficient, because it cannot settle the answer. It is resolvable in the sense of candidate-space reduction: the family name it does contain indexes a short, finite candidate set in a knowledge structure. Whether a system can then pick the right member is a separate question, which we also measure. When a model answers from such a page anyway, we call the behavior CLADE: category-level answering from deficient evidence. This is the point where a neural reader and a symbolic resource have the most to offer each other, and it is common: in our data, text names the exact disease for only a minority of graph facts, and part of the rest is written at the family level.

We ask three questions. How much of a biomedical knowledge graph does text name, and does a family name help a model at all? When a page names only the family, does a model stop declining, and are the answers it gives instead right? And can an ontology turn the family name back into the answer?

Our contributions are:

- We define resolvable insufficiency and the failure it invites, CLADE, and measure it per fact, by classifying each page as naming the answer, naming only its family in the MONDO disease ontology, or naming nothing.
- We show that, pooled over three 7–8B models from different families, a family-only page weakens abstention relative to a silent page, using a design that compares each question with itself with and without the page, and that the answers it frees are about as often wrong as right, far less reliable than answers from a page that names the disease.
- We isolate the cause with a counterfactual edit of the same pages, show that confidence thresholds, a larger model and an instruction to decline do not remove the effect,

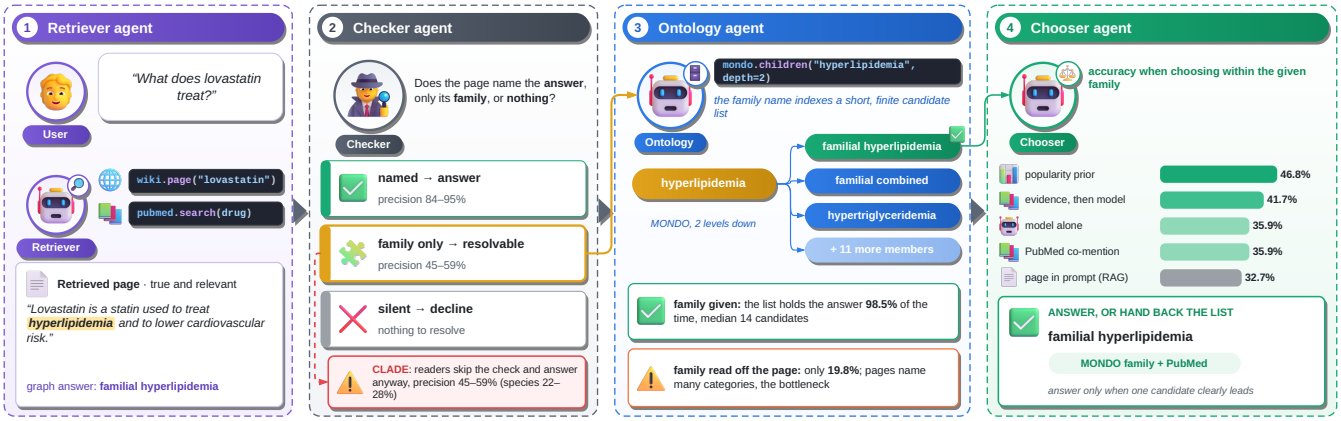


Figure 1: Resolvable insufficiency, drawn as a proposed agent workflow with the bottleneck we measure at each step. The dashed red path is CLADE: the reader answers from the family-only page instead of handing it on. The retrieved page is true and relevant but names only the answer's family. A page-type check should route it away from answering; an ontology step turns the family name into a short candidate list; a chooser picks from it or hands the list back. Only the ontology step works reliably today.

and replicate it on species interactions with a taxonomy.

- We decompose ontology-based recovery into finding the family, listing its members, and choosing among them, and show that the first step is the bottleneck, while a popularity prior is a strong baseline for the last.

2 Related Work

Insufficient context in RAG. Retrieval failures are a recognized failure point of RAG systems (Barnett et al. 2024). Joren et al. (2025) introduce sufficient context as a lens on RAG errors, label contexts as sufficient or insufficient, and show that models often answer correctly even from insufficient context and hallucinate rather than abstain. They note that a context can narrow the answer without settling it, but treat it as insufficient. Models are also known to be distracted by irrelevant context (Shi et al. 2023; Yoran et al. 2024), and Setiawan (2026) show that misleading passages collapse prompt-based abstention while missing passages do not. Our case differs from both: the page is neither irrelevant nor misleading. It is correct and points in the right direction, and we show that this alone is enough to weaken abstention.

Partial support and selective answering. Self-RAG (Asai et al. 2024) trains a model to judge whether a passage fully or partially supports its output, and ALCE (Gao et al. 2023) grades citations by whether they support a claim. CRAG (Yan et al. 2024) scores retrieved documents and searches again when they are weak, and SURE-RAG (Qiu, Han, and Huang 2026) verifies whether evidence is sufficient before answering. These methods recognize partial support, but not what kind of information is missing. A family name is a specific, structured kind of partial evidence: it says exactly which step is missing and where to look for it.

Abstention. Surveys and benchmarks document that models often answer when they should not (Wen et al. 2025; Kirichenko et al. 2025), including in medicine (Machcha et al. 2026), and propose methods to detect knowledge gaps (Feng et al. 2024; Kadavath et al. 2022). These works measure whether a model declines. We measure which kind of evidence makes it stop declining.

Ontologies and taxonomies in retrieval. OG-RAG (Sharma, Kumar, and Li 2024) grounds retrieval in a domain ontology, and TaSR-RAG (Sun et al. 2026) uses a taxonomy to structure multi-hop retrieval. The broader program of combining language models with knowledge graphs is surveyed by Pan et al. (2024), and neuro-symbolic AI more generally by d'Avila Garcez and Lamb (2023). These methods use symbolic structure for every query. We study the specific condition under which the page itself signals that a symbolic step is needed, and measure how far that step gets.

What text contains. Mallen et al. (2023) and Kandpal et al. (2023) relate model knowledge to entity popularity and to the number of relevant pretraining documents, and LMEnt (Gottesman et al. 2026) traces entity mentions through a pretraining corpus. These studies work at the level of entities. We measure, for each fact, at what level of specificity the text names it.

3 Setting and Data

Facts and ontology. We use drug–disease edges from PrimeKG (Chandak, Huang, and Zitnik 2023), whose indication and contraindication relations come from DrugCentral. Diseases are identified by MONDO (Vasilevsky et al. 2022), which we rebuild as a directed child-to-parent table of 17,801 edges over 11,788 diseases; 3,182 parent terms have two or more children.

Text. For each drug we use its English Wikipedia article and a set of PubMed abstracts retrieved per drug with a separate query budget for each relation. Wikipedia markup is cleaned before use: templates, tables, reference tags and file links are removed, and links are reduced to their visible text.

	Model knows it	Model does not
Text names it	5.2	6.2
Text does not	22.2	66.4

Table 1: Share of 1,800 PrimeKG facts (%) by whether Wikipedia names the disease and whether Qwen2.5 0.5B picks it from five options closed book.

How a page names a fact. For each edge we classify the drug's text into one of three groups. *Named*: the text contains the disease's exact name or an alternative name taken from Wikipedia redirects to the disease's article, which include synonyms, lay terms and abbreviations. *Family-only*: the text contains none of these but names one of the disease's MONDO ancestors within two levels. *Silent*: the text names neither. To separate real mentions from coincidence, each edge is paired with three diseases the drug has no relation to, and the same matcher is run on them.

4 How Much Does Text Reach?

We first measure, on 1,800 PrimeKG edges (600 per relation for indication, contraindication and off-label use), how often the drug's Wikipedia article names the disease and how often a model knows the fact closed book. The model picks the true disease from five options by mean token log-probability.

Table 1 crosses the two; here "names it" means the exact or an alternative name, so facts named only by their family count as not named. For two thirds of the facts, the text does not name the disease and the model does not know the answer. The "knows it anyway" cell is mostly chance: accuracy on facts the text does not name is 25.1% against a 20% chance rate, about 6% above chance after correction. A second corpus helps little: on a separate sample measured in both corpora, the share not named is 84.9% in Wikipedia, 81.1% in PubMed abstracts and 77.5% in both.

A family name alone does not help the model. If a family name carried useful signal, facts whose text names only the family would be easier for the model than facts whose text is silent. Testing this inside each relation and pooling with relation as a stratum on Qwen2.5 7B, family-only facts differ sharply from named facts (Mantel–Haenszel odds ratio 0.27, 95% CI 0.17–0.43, $p = 9 \times 10^{-9}$) and not measurably from silent facts (odds ratio 1.10, 95% CI 0.80–1.53, $p = 0.61$). The cleanest case is indications at 7B: 44.68% correct for family-only facts against 43.95% for silent facts. For contraindications the figures are 18.75% and 23.35%. An exact name raises the odds that the model knows a fact about 3.7-fold; a family name gives no measurable help on its own.

This motivates the rest of the paper. The family name is the most common form of partial evidence in this data, and the model cannot use it by itself. The question is what happens when such a page is placed in front of the model at answer time, and whether a symbolic step can make use of it.

5 CLADE: Abstention Under Family-Only Evidence

5.1 Design

Questions. We build 856 multiple-choice questions from indication and contraindication edges: 300 whose page names the answer, 256 whose page names only its family, and 300 whose page is silent (150 per relation and group, except 106 family-only contraindications, the most available). Each question offers five diseases and a sixth option, "Not enough information to tell," labeled A to F. Wrong options come from the answer's MONDO neighborhood: its siblings first, then its cousins (the other grandchildren of its grandparent), and random diseases only when those run out. Neighbors exclude the answer's own ancestors and descendants and any other disease the drug is related to by the same relation. Most wrong options thus share the answer's family, so a family name eliminates few of them.

Pages. Each question is asked twice: once closed book and once with a passage from the drug's text, its cleaned Wikipedia article followed by its PubMed abstracts. The passage is the first 800 characters of that text plus a 280-character window on each side of every mention of the disease or its family, merged where windows overlap and capped at 1,800 characters; for silent questions it is therefore the opening of the text. Edges whose cleaned passage no longer shows the answer or family are replaced by other edges.

Implementation. All models run on CPU with Hugging Face Transformers. Prompts are scored in batches with left padding and explicit position indices. For the recovery experiment, the prompt prefix shared by all candidates is encoded once and reused; a sample of these scores is checked against a full forward pass before each run. Code and data splits will be released.

Models and scoring. We test five instruction-tuned models: Qwen2.5 0.5B, 7B and 14B (Qwen Team 2024), OLMo 2 7B (Team OLMo et al. 2024), and Granite 3.1 8B (IBM Granite Team 2024). The prompt lists the options, and the model's answer is the option letter with the highest next-token probability, renormalized over the six letters. A question is *declined* when the model picks "Not enough information."

Controls. Two differences between the groups could produce a gap on their own. First, family-only questions more often have all four distractors from the neighborhood, which are harder to eliminate. We therefore analyze only such questions (106 named, 215 family-only and 160 silent). Second, the groups contain different facts, and some facts are harder than others even without a page. We therefore compare, for each question, its outcome with the page against its outcome closed book, and then compare this page effect between family-only and silent questions. The difference of these page effects is our main measure. Confidence intervals come from 5,000 bootstrap resamples of questions. To pool across the three 7–8B models, we average each question's page effect over the models before comparing groups, so each question counts once. The supplementary material gives the passage

Model	Page	Correct	Wrong	Declines	Prec.
Qwen2.5 7B	named	71.7	3.8	24.5	95.0
	family	26.5	18.1	55.3	59.4
	silent	13.8	2.5	83.8	84.6
OLMo 2 7B	named	77.4	15.1	7.5	83.7
	family	37.2	44.7	18.1	45.5
	silent	30.6	24.4	45.0	55.7
Granite 3.1 8B	named	72.6	7.5	19.8	90.6
	family	30.2	26.5	43.3	53.3
	silent	15.0	10.0	75.0	60.0
Qwen2.5 0.5B	named	77.4	14.2	8.5	84.5
	family	34.4	55.8	9.8	38.1
	silent	28.7	50.6	20.6	36.2
Qwen2.5 14B	named	67.9	2.8	29.2	96.0
	family	24.2	17.2	58.6	58.4
	silent	11.9	2.5	85.6	82.6

Table 2: Outcomes with the page (%), questions whose four distractors all come from the answer's neighborhood (106 named, 215 family-only, 160 silent). Precision is the share correct among questions the model answered.

and prompt formats, sampling details, formulas, queries and full result tables.

5.2 Results

With the page in hand, models decline less when it names only the family. Table 2 gives outcomes with the page. In all three 7–8B models, a family-only page leads to fewer declines than a silent page: 55.3% against 83.8% for Qwen2.5 7B, 18.1% against 45.0% for OLMo 2, and 43.3% against 75.0% for Granite. The model answers more, and it answers less precisely: among answered questions, family-only pages give 45.5–59.4% precision, against 83.7–95.0% for pages that name the disease.

The effect is the page's, not the questions'. The raw comparison also reflects differences between the facts in each group. With no page, family-only questions already produce more wrong answers than silent ones in all three models (32.6% against 15.6% for Qwen2.5 7B, 56.7% against 51.2% for OLMo 2, 33.0% against 26.2% for Granite; supplement), so part of the gap in Table 2 comes from which facts fall in which group. Figure 2 removes this by comparing each question with itself. Pooled over the three 7–8B models, a page raises declines by 14.7 points on family-only questions and by 28.5 points on silent ones, a difference of 13.8 points (95% CI -20.3 to -7.5, $p = 3.6 \times 10^{-5}$), which holds after a Bonferroni correction for the three outcomes ($p = 1.1 \times 10^{-4}$). The freed answers split about evenly. A page lowers wrong answers by 11.0 points on family-only questions and by 18.8 on silent ones, a difference of 7.7 points (95% CI 1.2 to 14.0, $p = 0.019$; 0.056 after correction), and it lowers correct answers by 3.7 and 9.8 points, a difference of 6.1 points (95% CI -0.1 to 12.4, $p = 0.058$). The decline effect is individually significant for OLMo 2 (-22.1 points, $p = 6 \times 10^{-6}$) and Granite (-18.2 points, $p = 0.0006$). Qwen2.5 7B shows no difference in its page effect (-1.1 points, $p = 0.82$): it already declines family-only questions less often without a page (32.1% against 59.4%), and the page raises declines by the same amount in both groups.

Robustness. The raw pattern holds for both relations in all three models (per-relation rates in the supplement), and the 0.5B model shows the decline effect as well (-8.4 points, 95% CI -15.7 to -1.1, $p = 0.027$).

The decline effect depends on how large the named family is. Splitting the family-only questions into thirds by the number of members, the pooled effect is -17.3 points (95% CI -26.2 to -8.2) for families of up to 10 members and -18.4 (-26.6 to -10.3) for 11–20 members, but only -6.5 (-15.7 to $+3.0$) for larger families (Figure 4c). A narrow family makes the page look answerable.

5.3 The Family Name Causes Clade

The comparison above is between different facts. To hold the fact fixed, we edit the page itself. For the 130 named questions whose answer has a parent in MONDO, we build two versions of each page: *family* replaces every mention of the answer with its parent's name, and *removed* deletes the sentences that mention it. Question, options and model are unchanged; only the page differs.

Figure 3 shows the result. Relative to the original page, the family version lowers correct answers by 21.0 points (95% CI -26.9 to -15.6) and raises wrong answers by 9.5 (5.9 to 13.6), while declines rise by only 11.5 (7.4 to 15.9). Relative to deleting the mention, keeping the family name lowers declines by 17.4 points (-23.6 to -11.5) and raises both wrong answers, by 5.1 points (0.5 to 10.3), and correct ones, by 12.3 (6.9 to 17.9). Each model shows the same order: Qwen2.5 7B declines 70.0% of the removed versions and 43.8% of the family versions, Granite 56.2% and 40.8%, OLMo 2 18.5% and 7.7%. The family name therefore produces both effects seen in the observational comparison: it suppresses declining, and the answers it frees split between right and wrong.

5.4 Three Easy Fixes Leave Clade in Place

Confidence does not flag these answers. A system could answer only when the model is confident. Figure 4a shows that confidence separates right from wrong answers less well on family-only pages than on named pages in every model (AUROC 0.59–0.72 against 0.73–0.87). The wrong answers are confident: the mean probability of a wrong choice on a family-only page is 0.77–0.99 for the 7B and larger models, and 0.99 for Qwen2.5 14B. A threshold that keeps most correct answers keeps most of these wrong ones too.

Scale does not remove the gap. Qwen2.5 14B is more precise on named pages than any smaller model (96.0%, Table 2), but on family-only pages its precision stays at 58.4%, and it gives 14.7 more points of wrong answers on family-only pages than on silent ones ($p = 6 \times 10^{-6}$). Like Qwen2.5 7B, it already declines family-only questions less often without a page, and its page effect does not differ between the groups ($+5.3$ points, $p = 0.29$).

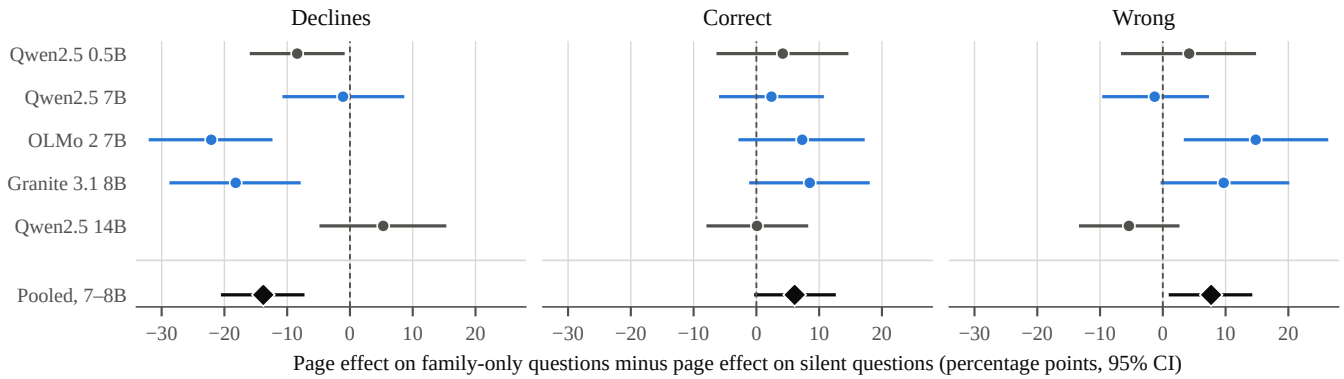


Figure 2: What a family-only page does, relative to a silent page. For each question, the page effect is its outcome with the page minus its outcome closed book; the plot shows the mean page effect on family-only questions minus that on silent questions, with 95% bootstrap intervals. The pooled row averages each question over the three 7–8B models (481 questions). Gray rows are outside the pooled set.

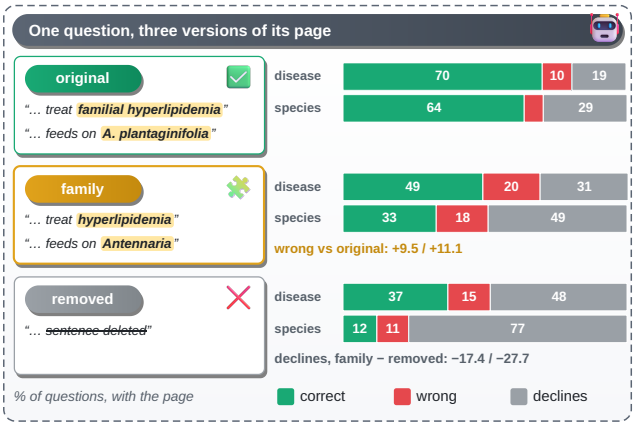


Figure 3: Counterfactual page edit on the same named questions, for diseases (PrimeKG, 130 questions) and species (GloBI, 255). *Family* replaces every mention of the answer with its parent; *removed* deletes the sentences that mention it. Outcomes with the page (%), averaged over three 7–8B models.

An instruction to decline is not targeted. The simplest remedy is to tell the model: if the page names only a broad category that contains the answer, choose "Not enough information." Figure 4b shows that the instruction raises declines on family-only pages by 15.2 points (95% CI 12.4 to 18.1), but also by 14.0 on silent pages and by 9.1 on pages that name the answer, where it costs 6.3 points of correct answers (−9.4 to −3.5). Precision on family-only pages stays at 47.9–64.9% after the instruction, far below the 89.6–94.7% the same prompt gives on named pages (per-model rates in the supplement). The instruction does not help because the models do not tell a family name from an answer.

What the result means. A silent page pushes a model toward "not enough information"; a page naming only the family pushes it much less, because it is relevant. The answers

	PrimeKG	GloBI
Questions, pooled (all-sibling)	481	833
Decline effect, family − silent	−13.8	−14.3
Wrong effect, family − silent	+7.7	+11.6
Precision, named page (%)	84–95	84–95
Precision, family-only page (%)	45–59	22–28
AUROC, named page	.73–.87	.82–.86
AUROC, family-only page	.66–.72	.52–.68
Edit: declines, family − removed	−17.4	−27.7
Edit: wrong, family − removed	+5.1	+6.7
Edit: correct, family − original	−21.0	−30.8

Table 3: The same tests on drug–disease facts (PrimeKG, MONDO) and on species interactions (GloBI, taxonomy), for the three 7–8B models. Effects in points; ranges are over models. All effects shown have 95% CIs that exclude zero.

this frees are about as often wrong as right: the family name lets the model guess within a smaller space without telling it which member is meant, and a system that treats a relevant page as permission to answer will do so at 45–59% precision while citing a relevant source.

5.5 CLADE in a Second Domain: Species Interactions

To test whether the effect depends on medicine or on MONDO, we repeat the abstention test and the counterfactual edit on species interactions from GloBI (Poelen, Simons, and Mungall 2014). We keep four relations (*eats*, *preys on*, *parasite of*, *pollinates*) between species, use the consumer's Wikipedia article as the page, and use GloBI's taxonomy (species, genus, family) in place of MONDO. A page is named if it gives the partner's binomial, its abbreviated form or a common name from Wikipedia redirects, and family-only if it gives only the genus or family. Wrong options are species of the same genus, then of the same family, and never a recorded partner. The question set has 75 questions per group and relation (900 in all).

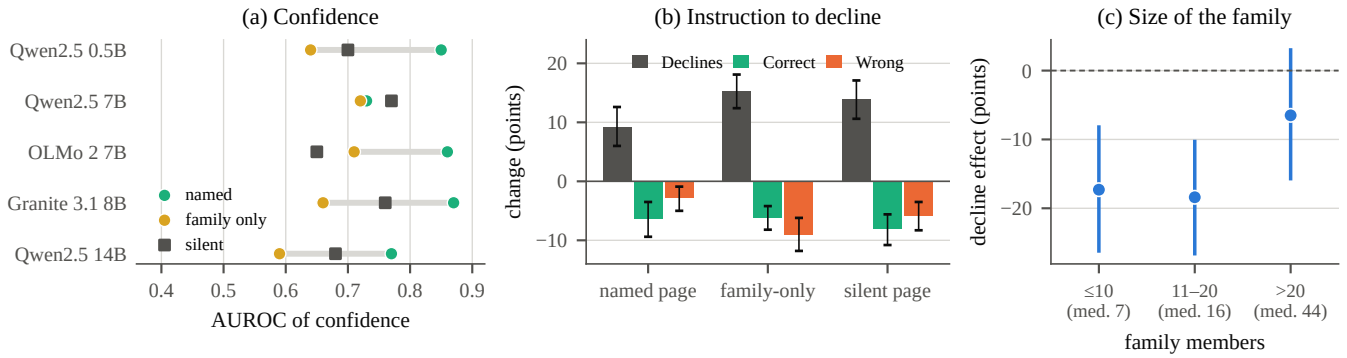


Figure 4: Three easy fixes. (a) Confidence separates right from wrong answers less well on family-only pages in every model. (b) An instruction to decline when the page names only a broad category raises declines on every kind of page (pooled over three 7–8B models, 95% CIs). (c) The decline effect of a family-only page is concentrated in small and medium families.

Exact partners are rarely written down: of 23,694 measured edges, the page names the partner species for 3.9% and only its genus or family for 7.1%. Table 3 shows that every result carries over. A page raises declines by 12.1 points on silent questions but lowers them by 2.3 on family-only ones, a difference of -14.3 points ($p = 1.4 \times 10^{-9}$); the freed answers are mostly wrong ($+11.6$ points, $p = 2.5 \times 10^{-8}$; correct $+2.7$, $p = 0.09$). Family-only precision falls to 22–28%, since species of one genus are harder to tell apart than diseases of one family. The counterfactual edit is stronger than on PrimeKG: keeping the genus name instead of deleting the sentence lowers declines by 27.7 points (-32.2 to -23.5) and raises wrong answers by 6.7 (3.4 to 10.2; Figure 3). Resolvable insufficiency is a property of how text names facts that sit in a hierarchy, not of one domain.

6 Resolving the Insufficiency with an Ontology

If a family name is the most common form of partial evidence, an ontology should be able to use it. We decompose this into three steps: identify the family from the page, list its members, and choose among them.

6.1 Design

We use the 410 indication and contraindication edges whose text names only the family, in Wikipedia and PubMed together. A pick counts as correct when it is any disease the drug has the same relation to, since a family list holds 2.47 true answers on average and over half hold more than one.

Listing the members. The *given-family* list takes the narrowest ancestor of the answer that the text names and lists its MONDO descendants up to two levels below it, capped at 64. It assumes the right family is known. The *page-built* list instead finds every MONDO category with at least two children whose name appears in the drug's Wikipedia article, and takes the union of their descendants. A list larger than 64 candidates counts as too broad.

Choosing a member. We compare Qwen2.5 7B choosing from the given-family list without the page and with it; Wikipedia evidence (the candidate's own article names the drug); PubMed evidence (the number of abstracts naming both the drug and the candidate, with a relation-specific qualifier); PubMed evidence normalized by the candidate's total number of abstracts; and a popularity prior, the candidate with the most PubMed abstracts regardless of the drug. Two baselines use no ontology: a list of the same size with the answer and random diseases, and the same list with the page in the prompt, which is what a standard RAG system sees. Paired comparisons use McNemar's test (McNemar 1947).

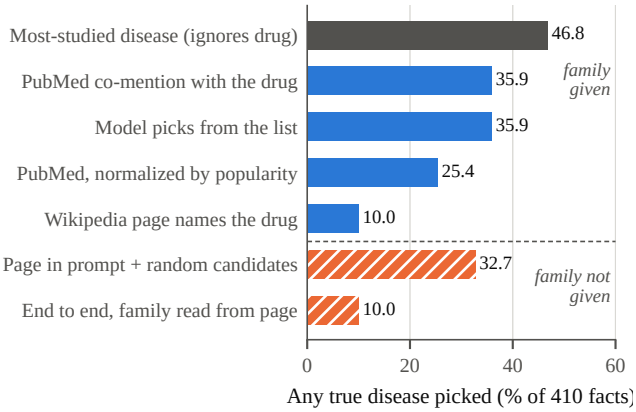


Figure 5: Recovery on 410 family-only facts. Blue and gray bars choose within the given family; hatched bars do not receive the family.

6.2 Results

Given the family, the ontology narrows the search well. Table 4 shows that when the right family is known, its member list contains the answer 98.5% of the time with a median of 14 candidates. List coverage is not accuracy, however: a member still has to be chosen.

Finding the family is the bottleneck. When the list is built from the categories the page actually names, it contains the answer for only 19.8% of facts. A page names a median of

	Given family	Page-built
Usable lists (of 410)	410	166
List contains the sampled answer	98.5%	19.8%
List contains any true answer	99.5%	24.6%
Median list size	14	32

Table 4: Candidate lists. Coverage is over all 410 facts; of the 244 unusable page-built lists, 209 exceed 64 candidates and 35 have no usable category.

five categories, and for 209 of 410 facts their union exceeds 64 candidates. The page does name the right family, but it also names others, and nothing in the text marks which one belongs to the answer. Reading more text makes this worse, not better. Building the list from the Wikipedia article and the PubMed abstracts together raises the median number of categories named to 17; 391 of 410 lists then exceed 64 candidates, and only 3.7% of facts get a usable list that contains the answer. The bottleneck is not missing text but too many candidate families.

Within a family, popularity beats evidence. Figure 5 compares ways of choosing. Given the family, the model alone is right 35.9% of the time, and 32.4% when it also sees the page. PubMed co-mention also reaches 35.9%, and taking evidence first (Wikipedia and PubMed combined) and the model's pick otherwise reaches 41.7%, which beats the model alone (59 against 35 discordant facts, $p = 0.018$). The popularity prior, which ignores the drug, reaches 46.8%, the highest of all. Once co-mention counts are divided by the candidate's total abstracts, they fall to 25.4%, so the drug-specific part of the literature signal is weak. Wikipedia rarely helps, because a candidate's article names the drug in only 3.9% of cases. The two baselines without an ontology reach 34.9% (random list) and 32.7% (page in prompt). Their lists exclude the drug's other true partners, so under any-true-answer scoring they are not at matched chance with the family lists and serve as reference points only.

What the misses look like. Some misses choose a better-studied member of the family (for carboplatin, ovarian carcinoma rather than the rarer germ cell tumor the graph records), and some are true facts the graph does not list (moxifloxacin for Legionnaires' disease), so evaluation against a graph understates recovery for every method alike. The supplement lists examples.

End to end. Without being given the family, building the list from the page and choosing by evidence recovers the answer for 10.0% of facts. The page-in-prompt baseline reaches 32.7%, helped by a family name that lets it eliminate random distractors outside the family. With lists built from both corpora, end-to-end recovery falls to 1.5%, for the reason given above.

7 Discussion

Where the neural reader should hand over. The results give a precise handover point. Under resolvable insufficiency, a model reading a family-only page shows CLADE: it treats the page as license to answer, with markedly lower precision than on named facts. A symbolic structure can turn the same family name into a short list that almost always contains the answer, but only if the right family is identified. A neuro-symbolic system should therefore do three things the models do not: recognize that a page names a family rather than the answer, identify which family belongs to the question, and choose within it with a baseline that is at least as strong as popularity.

A procedure that follows from the results. The findings suggest a simple decision procedure for a retrieval-augmented system facing a page about the question's subject.

1. Check whether the page names the answer type at the required level. If it does, answer from the page.
2. If the page names only categories, identify which category belongs to the question. If none can be identified, decline.
3. List the category's members from the ontology and score them with evidence and with a popularity prior; answer only if a candidate clearly leads, and otherwise return the short list or decline.

The models we tested skip the first two checks, which is CLADE: a relevant page is treated as permission to answer, and neither confidence nor an instruction recovers step 1. Step 2 is the weakest link in our measurements and the most promising target for a learned component. For step 3, any system should report a popularity prior: in our data it beats the model, literature counts and a standard RAG baseline.

Partial evidence has a type. Sufficient-context methods ask whether a page settles the answer. Our results argue for also asking how it fails to. A page that names the family fails differently from one that is silent or misleading: it carries usable information, it weakens abstention, and it offers a symbolic route to narrowing the answer. Labeling this case separately would let systems route it to an ontology rather than either answering or declining.

8 Limitations

The study covers two domains, drug–disease facts with MONDO and species interactions with a taxonomy; recovery is evaluated on the first only. Questions are multiple choice with a fixed set of options, so abstention is measured as choosing "Not enough information" rather than as free-text refusal. The models are open and at most 14B, and the counterfactual edit covers only named questions whose answer has a MONDO parent. Recovery is evaluated with one model, and the page-built list uses exact category-name matching, which a learned family classifier could improve. The PrimeKG family-only group is small for contraindications (106 questions).

9 Conclusion

Models meet resolvable insufficiency with CLADE: given a page that names only the answer's family, they stop declining and answer far less reliably, in medicine and in ecology. An ontology lists the answer once the family is known; finding the family and choosing within it remain open.

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